# CLI GUIDE

# HYPERCLUSTER CLI GUIDE

Task-oriented reference for the bash CLI. Generated CLI help is also at docs/_build/cli-reference.txt (run make docs).

## Quick paths

Goal · Commands
**Remote k3s platform** · preflight ? deploy ? test
**Kubespray HA cluster** · wizard or edit conf ? preflight ? deploy
**Automation / CI** · preflight --json, test --json, health --json, status --json
**Desktop / inventory** · discover --json, vms --json, workloads --json
**Day-2 add nodes** · Update WORKER_IPS ? scale
**Upgrade K8s** · Bump KUBE_VERSION ? upgrade

## Global flags

Flag · Meaning
-c <file> · Config file (default ./cluster.conf)
-f · Force inventory regen; skip destroy confirm; force remove-node
-n · Dry run
-v · Verbose Ansible output
-V · Print version
-h · Help

## Commands

Command · k3s · Kubespray · Description
wizard · ? · ? · Interactive cluster.conf generator
init --profile <name> · ? · ? · Copy config/profiles/<name>.conf.example ? cluster.conf
init · ? · ? · Clone Kubespray + inventory
preflight · ? · ? · Local tools + SSH checks
preflight --json · ? · ? · Machine-readable preflight JSON
deploy · ? · ? · Full install
test · ? · ? · Platform smoke tests
test --full · ? · ? · Extended snapshot checks
test --json · ? · ? · Machine-readable results
status · ? · ? · Node status
status --json · ? · ? · Machine-readable node JSON
health · ? · ? · Connect + unified report (nodes, pods, services, VMs)
health --json · ? · ? · Machine-readable health report
health --with-test --json · ? · ? · Include platform smoke in report (k3s)
discover --json · ? · ? · Discover hosts/VMs (no config required)
vms --json · ? · ? · KubeVirt VMIs + template catalog

vm create --json · ? · ? · Validate template; create stub (preview)
workloads --json · ? · ? · Pod and service inventory
scale · partial · ? · Add workers from conf
upgrade · ? · ? · Upgrade to KUBE_VERSION
remove-node · ? · ? · Drain + remove node
fetch-kubeconfig · ? · ? · Copy admin kubeconfig
merge-kubeconfig · ? · ? · Merge into ~/.kube/config
install-kubevirt · ? · ? · KubeVirt + CDI
destroy · ? · ? · Tear down cluster
unlock · ? · ? · Clear stale lock

## k3s platform workflow

cp config/cluster.conf.example cluster.conf
**# or:**

```
./hypercluster init --profile k3s-remote
```

$EDITOR cluster.conf  # K8S_DISTRO=k3s, METALLB_IP_POOL, SSH_*, CONTROL_PLANE_IPS

```
./hypercluster -c cluster.conf preflight
./hypercluster -c cluster.conf preflight --json
./hypercluster -c cluster.conf deploy
./hypercluster -c cluster.conf test
./hypercluster -c cluster.conf health --json
./hypercluster -c cluster.conf status --json
./hypercluster -c cluster.conf workloads --json
```

**Deploy behavior:** skips k3s reinstall when server is healthy. Set FORCE_K3S_REINSTALL=1 to replace.
**Stack order:** Cilium ? cert-manager ? MetalLB ? KubeVirt/CDI ? metrics-server.
**Per-node SSH (k3s):** set NODE_SSH_OVERRIDES_JSON to a JSON array of {ip, sshUser?, sshKey?, sshPort?} objects when nodes need different credentials than global SSH_*.

## Discover (no cluster.conf)

# CSV fixture (unit test / import preview)

```
./hypercluster discover --json --source csv --csv tests/fixtures/discover-nodes.csv
```

# Other sources: libvirt, proxmox, vmware, ssh-scan (see lib/discover.sh)

```
./hypercluster discover --json --source libvirt --uri qemu:///system
```

## JSON output

**`test --json`:**
**Stdout (stderr has human PASS/FAIL lines):**

{"pass": 10, "fail": 0, "checks": [{"name": "Cilium healthy", "passed": true}]}

**`status --json`:**

{"ready": 1, "total": 1, "nodes": [{"name": "node1", "ready": true, "roles": "true"}]}

**`health --json`:**

Unified connect + report (stdout JSON only; human summary on stderr without --json):

{
 "cluster": "k3s-175",
 "connected": true,
 "state": "healthy",
 "issues": [],
 "status": {"ready": 1, "total": 1, "nodes": []},
 "workloads": {"pods": [], "services": []},
 "vms": {"vms": [], "templates": []},
 "summary": {
  "nodes_ready": 1, "nodes_total": 1,
  "pods_running": 66, "pods_total": 67,
  "pods_pending": 0, "pods_failed": 0,
  "services": 55, "vms_running": 0, "vms_total": 0
 },
 "tests": null
}

state is healthy, degraded, or unreachable. Add --with-test on k3s to populate tests from platform smoke.

**`workloads --json`:**

{"pods": [{"namespace": "kube-system", "name": "cilium-abc", "phase": "Running", "node": "10.0.0.10", "ready": 1, "containers": 1}], "services": [{"namespace": "default", "name": "kubernetes", "type": "ClusterIP", "clusterIp": "10.43.0.1"}]}

**`vms --json`:**

{"vms": [], "templates": [{"id": "ubuntu-22.04", "label": "Ubuntu 22.04 LTS", "os": "linux"}]}

**`vm create --json`:**

{"status": "stub", "message": "VM create from template is not yet implemented ? catalog entry validated", "template": {}, "name": "my-vm", "namespace": "default"}

## Make shortcuts

```
make deploy CONFIG=cluster.conf
make test-remote CONFIG=cluster.conf
make status CONFIG=cluster.conf
make desktop-check    # TS build + discover/vms/workloads JSON tests
make docs
```

See docs/generated/MAKE-TARGETS.md for full list.

## Related

- Desktop UX guide
- Configuration reference
- Documentation hub

_Doc version: 2026-06-17T17:22:39Z · hypercluster v0.1.0_